Psychology and Neuroscience 37th Annual Graham Goddard In-House Conference

April 28/29, 2011
Life Sciences Centre
Room 4258/4263

The 2011 Psychology and Neuroscience Graham Goddard In-House Conference

Welcome! We would like to thank you for attending the 37th Annual Graham Goddard In-House Conference. This is an excellent opportunity to hear about the ongoing research of students, faculty and colleagues. It is your participation that makes this event a success.

Thanks should also be extended to the In-House Conference Committee. The time that the following individuals put into the planning and organization of the In-House is greatly appreciated: Richard Brown, Sophie Jacques, Suzanne King, Jeffrey MacLeod, Kyle Roddick, Emily LeDue, Amanda Hudson, Kurt Stover and Melissa Stewart.

Dr. Graham V. Goddard

Dr. Graham V. Goddard (1938-1987) was a leading researcher at Dalhousie University. Goddard invented the kindling model of epilepsy that has, since its introduction in the 1960s, become widely used research tool in studies of long-term potentiation (LTP) and epilepsy. His work was published in Nature, Science, along with other high profile journals. During his time at Dalhousie, Goddard helped establish Dalhousie as a leading research institution. Goddard also organized the annual Psychology Department In-House Conference, to help investigators in the department update others about their work and encourage academic dialogue. It is in his honour that we continue this tradition today.

Friendly Reminders

Talks will be held in LSC 4253/4268.
Coffee and tea will be available in the 4th Floor Study Centre.
Lunches will take place in the 2nd Floor Lounge.
There will be a dinner and reception to follow at the end of Friday April 29th, in the 2nd Floor Lounge.

Schedule at a Glance

Thursday April 28th, 2011

8:30 – 10:30	**Session 1 Chair: Amanda Hudson**		
	8:30	T1)	Introduction and Brief History of Convention *Raymond Klein & Richard Brown*
	9:00	T2)	The Impact of the sleep lab environment on sleep quality in children with attention-deficit/hyperactivity disorder (ADHD) and typically developing peers *Jennifer Richards & Penny Corkum*
	9:15	T3)	The Effect of Cognitive Load on Prism Adaptation *Sarah Vinette & Gail Eskes*
	9:30	T4)	The Development of Preferences: Investigating Age Differences in the Mere Exposure Effect *Michelle Kerr, Jillian Filliter & Shannon Johnson*
	9:45	T5)	The Role of Attentional Control Settings in Inhibition of Return *Matthew Hilchey & Raymond Klein*
	10:00	T6)	Improving Cognitive Function in Healthy Aging *Gail Eskes*
	10:15	T7)	Isolating Contributions of Voltage Gated Sodium Channel Isoforms to the Electroretinogram *Ben Smith, Patrice Cote & Francois Tremblay*
10:30 – 11:00	**Coffee Break – LSC 4th Floor Study Centre**		
11:00 – 12:30	**Session 2 Chair: Kurt Stover**		
	11:00	T8)	The Use of Olfactometry to Assess Learning and Memory in Mouse Models of Alzhiemer's Disease *Kyle Roddick & Heather Schellinck*
	11:15	T9)	Investigation of Cytoskeletal Proteins in Neurons of the Cat Lateral Geniculate Nucleus *Emily LeDue, Nathan Crowder & Kevin Duffy*
	11:30	T10)	Can You Tell if Kids Have Pain or Not? Detecting Deception in Children's Facial Expression of Pain *Katelynn Boerner, Christine Chambers, Ken Craig & Rebecca Pillai-Riddell*

11:45 T11) Physiological Arousal in Children and Adults with an Autism Spectrum Disorder: Reliability, Individual Differences, and the Role of Context
Dörthe Haacker, Shannon Johnson, Vickie Armstrong, Susan Bryson & Jillian Filliter

12:00 T12) Speed-Accuracy Tradeoff in Stroop Interference
Virginia Palango & Jason Ivanoff

12:15 T13) Effects of Training on a Dual n-Back Task on the Components of Working Memory and Fluid Intelligence
Andrew Clouter & Gail Eskes

12:30 – 1:30 **Lunch and Poster Session – 2nd Floor Lounge**

P1) Personality Factors and Susceptibility to Burnout in Nova Scotia Ground Ambulance Paramedics: A Cross-Sectional Study
Zach Dewar, Alix Carter & Gail Eskes

P2) headgames.org: A Secure Platform for Web-Based Experiments
Mike Lawrence & John Christie

P3) Epigenetics and the Future of Psychology and Neuroscience
Richard Brown

P4) A Neurodevelopmental Profile of the 3xTg-AD Mouse Model of Alzheimer's Disease
Caitlin Blaney and Richard Brown

P5) Orientation Specificity of Contrast Adaptation in Rodent Primary Visual Cortex
Aaron Stroud, Emily LeDue & Nathan Crowder

P6) Is Cognitive Performance Associated with Preclinical Markers of Parkinson's Disease?
Laura Burns, Kim Good, John Fisk & Harold Robertson

P7) Modeling Inhibition of Return as Short-Term Depression of Early Sensory Input to the Superior Colliculus
Jason Satel, Zhiguo Wang, Thomas Trappenberg & Raymond Klein

1:30 – 3:15 **Session 3 Chair: Emily LeDue**

1:30 T14) The Emergence of Imitation: Preliminary Findings from a Prospective Study of Younger Siblings of Children with Autistic Spectrum Disorders
Ainsley Boudreau, Isabel Smith, J. Brian, Susan Bryson, W. Roberts, C Roncadin, P. Szatmari & L. Zwaigenbaum

1:45	T15)	Repent Your Statistical Sins *Mike Lawrence*
2:00	T16)	Immune Activation Influences Agonistic Behaviour in the Texas Field Cricket *Gryllus texensis* *Evan Fairn & Shelley Adamo*
2:15	T17)	The Remarkable Story of Captain Trevor Greene *Ryan D'Arcy, Stephen Lindsay, Debbie Lepore & Trevor Greene*
2:30	T18)	Repertoire Composition, Song Structure and Singing Behaviour in the Hermit Thrush *Sean Roach, Lynn Johnson & Leslie Phillmore*
2:45	T19)	Effects of Gambling-Related Cues on Automatic Activation of Gambling Outcome Expectancies *Melissa Stewart & Sherry Stewart*
3:00	T20)	A Checklist for Documenting Laboratory Factors Affecting the Behavioral Phenotyping of Mice *Richard Brown, Timothy O'Leary, Rhian Gunn & Heather Schellinck*
3:15 – 3:45	**Coffee Break – LSC 4th Floor Study Centre**	
3:45 – 6:00	**Session 4 Chair: Aaron Stroud**	
3:45	T21)	Learning, Memory and Social Behaviour in the 3xTG-AD Mouse Model of Alzheimer's Disease *Kurt Stover & Richard Brown*
4:00	T22)	The Perfectionism Model of Binge Eating: A Self-Perpetuating Cycle *Sean Mackinnon, Simon Sherry, Aislin Graham, Sherry Stewart, Daba Sherry, Stephanie Allen, Skye Fitzpatrick & Dan McGrath*
4:15	T23)	The Relationship between Empathy and the Visual Scanning of Emotional Faces *Dana Higginbotham, Shannon Johnson, Jillian Filliter & Heath Matheson*
4:30	T24)	Making Working Memory Better: Training the Central Executive *Ryan Wilson, Stefan Allen & Gail Eskes*
4:45	T25)	Working Working Memory *Stefan Allen, Ryan Wilson & Gail Eskes*
5:00	T26)	The Effects of Repeated Stress in Adolescence on Sensorimotor Gating, Affect, and Memory in Male and Female Rats *Namrata Joshi & Tara Perrot*

5:15 T27) There is a Coyote in my Backyard!
Simon Gadbois

5:30 T28) Examining the Video Deficit Effect in 12 – 42 Month Olds
Oriane Landry & the Students of PSYO 3091

5:45 T29) Characterizing the Hemodynamic Response Function of White Matter Using 4T fMRI
Leanne Fraser, T. Stevens, C. Liu & Ryan D'Arcy

Friday April 29th, 2011

8:30 – 10:15 **Session 5 Chair: Kyle Roddick**

8:30 T30) Investigating Sex Differences in Face Recognition and Gaze Direction
Sarah Whitzman, Laura Goodman, Heather Phelan & Shannon Johnson

8:45 T31) Factors Influencing Card-Sorting Performance in 3-year-olds
Denise Lewis, Meghan Tower & Oriane Landry

9:00 T32) It's All in the Expression: Cricket Tales
Russell Easy & Shelley Adamo

9:15 T33) Put on a Happy Face! Relating Emotional Competence and Emotion Regulation
Amanda Hudson & Sophie Jacques

9:30 T34) Introducing an Endogenous Attention Systems Test
Mike Lawrence & Shannon Johnson

9:45 T35) Human Sensitivity to Auditory Temporal Irregularity
Moragh Jang, Rachel Dingle, Susan Hall & Dennis Phillips

10:00 T36) Sensitivity to Morphological and Orthographic Stress Cues When Reading Multisyllabic English Words
Erin Sparks & Helene Deacon

10:15 – 10:45 **Coffee Break – LSC 4th Floor Study Centre**

10:45 – 12:15 **Session 6 Chair: Jeffrey MacLeod**

10:45 T37) Visual and Auditory Factors in Sound Localization Acuity
Dennis Phillips, Chelsea Quinlan & Rachel Dingle

11:00 T38) Exploring Attention in Item-Method Directed Forgetting
Kate Thompson & Tracy Taylor

11:15 T39) The Relationship Between Inflammation and Neurotrophin Function in Depression
Cai Song

11:30 T40) Is tobacco addiction nicotine addiction?
Sean Barrett

11:45 T41) Spatial and Temporal Dynamics of Visual Attention
Yoko Ishigami & Raymond Klein

12:00 T42) Validating a Measure of Parental Sleep Beliefs in the School-Aged Population
Meredith Bessey, Aimee Coulombe & Penny Corkum

12:15 – 1:15 **Lunch and Poster Session – 2nd Floor Lounge**

P8) The Effect of the Pulfrich Stereophenomenon on Reaching and Perceptual Judgment
Stéphanie Beckett & David Westwood

P9) Visuo-Spatial Learning and Memory in Six Mouse Models of Alzheimer Disease
Richard Brown, Rhian Gunn & Timothy O'Leary

P10) Neural and Behavioural Analyses of the 5XFAD Mouse model of Alzheimer's Disease
Dietmar Hestermann, Timothy O'Leary, Arjan Blokland & Richard Brown

P11) Exploring the Modulation of Attentional Capture by Spatial Attentional Control Settings: Converging Evidence from Event-Related Potentials
Yoko Ishigami, Jeff Hamm, Jason Satel & Raymond Klein

P12) Changes in Descending Noradrenergic Pain Pathways Following Sciatic Nerve Injury
Jean Liu, Jana Sawynok, Gary Allen & Allison Reid

P13) Modeling Mechanisms of Reversed Effects of Manipulability on Naming and Categorizing Objects
Joshua Salmon, Patricia McMullen & Thomas Trappenberg

P14) The Localization of Neurofilament Subunits in the Rat Primary Somatosensory Cortex Throughout Postnatal Development
Lin Lin Ngu, Song, Katharine Cheung & Kevin Duffy

1:15 – 3:15 **Session 7 Chair: Melissa Stewart**

1:15 T43) Development of Intrinsic Neurons in the Mushroom Bodies of *Drosophila melanogaster*

Katie Goodine, Nancy Butcher & Ian Meinertzhagen

1:30 T44) The Effects of Alcohol on Smoking Cue Reactivity
Marcel Peloquin

1:45 T45) Autism as a Cognitive Style: Visual Orienting and the Broader Autism Phenotype
Oriane Landry

2:00 T46) Testing the Embodied Account of Object Recognition
Heath Matheson & Patricia McMullen

2:15 T47) Sensory and Motor Mechanisms of Oculomotor Inhibition of Return
Zhiguo Wang, Jason Satel & Raymond Klein

2:30 T48) Sleep Architecture in Children with ADHD and Their Typically Developing Peers
Andre Benoit & Penny Corkum

2:45 T49) Odour Matching in Dogs: Applications and Challenges
Simon Gadbois, Kate Thompson, Brittany Hilton, John Christie & Vincent LoLordo

3:00 T50) The Role of Faces in Item-Method Directed Forgetting
Chelsea Quinlan & Tracy Taylor

3:15 – 3:45 **Coffee Break – LSC 4th Floor Study Centre**

3:45 – 6:00 **Session 8 Chair: Kyle Roddick**

3:45 T51) Wait a Minute, Forget That: Using Natural Speech Cues to Induce Directed Forgetting
Ben Corrigan, Jonathan Fawcett & Tracy Helmick

4:00 T52) Drinking Motives, Alcohol Use, and Alcohol-Related Problems: A Cross-National Comparison and Validation of the Drinking Motives Questionnaire-Revised Among College Students
Marie-Eve Couture, Sherry Stewart, E. Kuntsche, M. Cooper & Roisin O'Connor

4:15 T53) Aladdin's Cave: New Lamps for Old in Illuminating the Tiny World of the Fly's Eye
Steve Shaw

4:30 T54) The Effects of Prenatal Psychological Stress and Postnatal Environmental Enrichment on the Development of Anxiety and Depressive Behaviours and Neural Changes in the HPA Axis
Amanda Green, Kal Mungovan, Meaghan MacLean & Tara Perrot

4:45 T55) Reading for Meaning: Investigating Children's Sensitivity to Morphological Structure During Contextual Reading
Kyle Levesque & Hélène Deacon

5:00 T56) White Matter Connectivity in Bilaterally Deaf Individuals vs. Hearing Controls
Therese Chevalier, Aaron Newman, Vanessa Lim, Heather Maessen & Manohar Bance

5:15 T57) Motivation and Data Quality Across the Term: How to Get Cleaner Data in Just Five Minutes
Jonathan Fawcett, Heath Matheson & John Christie

5:30 T58) The Effects of Eating Dennettia tripetala fruit on locomotor and anxiety-related behaviours in CD1 mice
Daniel Ikpi, Richard Brown & Sunday Bisong

5:45 Closing Remarks
Raymond Klein

6:00 Dinner and Reception – 2nd Floor Lounge

T1) Introduction and Brief History of Convention
Raymond Klein & Richard Brown

T2) The Impact of the sleep lab environment on sleep quality in children with attention-deficit/hyperactivity disorder (ADHD) and typically developing peers
Jennifer Richards & Penny Corkum

Children with ADHD have significant sleep problems relative to their typically developing (TD) peers. Research in this area has yielded inconsistent findings pertaining to sleep quality in children with ADHD. This could be due to the fact that research often uses polysomnography (PSG) to measure sleep, yet it is unknown if PSG data collected in a sleep lab is representative of children's typical sleep at home. This study examined whether children with ADHD's sleep quality was differentially affected by the sleep lab environment relative to TD children by collecting actigraphy data in the home environment and in a sleep lab. Data from 30 sex-matched and age-matched children, aged 6-12 was analyzed. Results revealed that some sleep quality parameters were differentially impacted by the sleep lab among children with ADHD. Results also demonstrated that, irrespective of an ADHD diagnosis, sleep efficiency was better in the sleep lab compared to at home.

T3) The Effect of Cognitive Load on Prism Adaptation
Sarah Vinette & Gail Eskes

Prism adaptation (PA) has been found to ameliorate symptoms of visuo-spatial neglect, a deficit in orienting one's attention to the left side of space arising from damage to the right hemisphere of the brain. PA is the process of adapting visuo-motor function while wearing vision-displacing prismatic goggles. Pointing performance is measured with various tests before, during and after an adaptation (exposure) period where goggles are worn. The difference in pointing accuracy between the pre-test and post-test is the aftereffect. Cognitive load during the exposure period has been reported to affect exposure performance and final after-effect due to PA, although the findings are inconsistent and may depend upon the method of adaptation. This study examined the impact of cognitive load on exposure performance and aftereffects using two methods of adaptation that varied in amount of cognitive control needed for pointing. Adaptation appeared less affected by need for cognitive control than expected.

T4) The Development of Preferences: Investigating Age Differences in the Mere Exposure Effect
Michelle Kerr, Jillian Filliter & Shannon Johnson

The development of preferences is an emotional process that allows us to form values, commitments, and attachments. One of the simplest forms of preference acquisition is demonstrated by the mere exposure effect (MEE; Zajonc, 1968); the repeated exposure of a stimulus is sufficient for enhancing an individual's attitude toward it. Although research has consistently demonstrated a MEE in adults, children have yielded inconsistent findings (Houston-Price, 2009), and it has been suggested that children prefer novel rather than familiar stimuli (i.e. a *reverse* MEE; Bornstein, 1989). To examine this emerging preference for familiarity, we have taken a lab-based approach to investigate the MEE in 30 typically developing youth aged 8 to 14 years. As hypothesized, the strength of the MEE was more pronounced in older children. The findings have interesting implications regarding the development of implicit memory systems, and the influence of parenting styles and media advertisements within the child population.

T5) Are there bilingual advantages on non-linguistic interference tasks?

Matthew Hilchey & Raymond Klein

It has been proposed that the unique need for early bilinguals to manage multiple languages while their executive control mechanisms are developing might result in long-term cognitive advantages on inhibitory control processes that generalize beyond the language domain. We review the empirical data from the literature using non-linguistic interference tasks to assess the validity of this bilingual inhibitory control advantage (BICA) proposal. Our review of these findings reveals that the bilingual advantage on conflict resolution, which by hypothesis is mediated by inhibitory control, is sporadic at best, and in some cases conspicuously absent. A robust finding from this review is that bilinguals typically outperform monolinguals on both compatible and incompatible trials. Together, these findings suggest that bilinguals enjoy a more widespread cognitive advantage (bilingual executive processing advantage, BEPA) that is likely observable on a variety of cognitive assessment tools but not on traditional assays of non-linguistic inhibitory control processes.

T6) Improving Cognitive Function in Healthy Aging

Gail Eskes

Exercise can improve cognitive abilities in healthy older adults and delay the onset of age-related cognitive decline. The mechanisms for the positive benefit of exercise and how these effects interact with other variables known to improve cognitive function (e.g., involvement in cognitive activities) were examined in 42 healthy post-menopausal women. The association of physical fitness (VO_2 max), cerebral blood flow and cognitive activity with neuropsychological performance was assessed through linear regression. Physical fitness, cerebrovascular reserve and total number of cognitive activities (but not total hours) were independent predictors of cognitive function. In addition, prediction of neuropsychological performance was better with multiple variables than each alone. Cognitive function in older adults is associated with multiple factors, including physical fitness, cerebrovascular health and cognitive stimulation. Interestingly, cognitive stimulation effects appear related more to the diversity of activities, rather than the duration of activity. Further examination of these relationships is ongoing in a prospective cohort study.

T7) Isolating Contributions of Voltage Gated Sodium Channel Isoforms to the Electroretinogram

Ben Smith, Patrice Cote & Francois Tremblay

10 Na_v channel isoforms with differing properties contribute to global sodium influx but pharmacologically only two groups can be isolated. We have applied a series of Na_v channel antibodies conjugated to an ampipathic peptide to isolate the contributions of individual isoforms to the summed electrical potential of the retina. Both pan Na_v antibodies and a cocktail of antibodies against the 3 most common Na_v channels in the retina (Na_v1.1, 1.2, and 1.6) have similar effects on low frequency component of the electroretinogram (ERG). These antibodies replicate the effects of tetrodotoxin, the canonical Na_v channel blocker. Individual Na_v1.6, 1.2, and 1.1 antibodies indicate that Na_v1.6 and to a lesser extent Na_v1.2 carry the majority of voltage gated sodium current in neurons contributing to the ERG. Interestingly we found that both the pan Na_v antibody and an antibody against the TTX resistant Na_v channel 1.8 reduce the high frequency components of the ERG.

T8) The Use of Olfactometry to Assess Learning and Memory in Mouse Models of Alzhiemer's Disease

Kyle Roddick & Heather Schellinck

Alzheimer's disease is a progressive, neurodegenerative disease that affects half a million Canadians. Patients with Alzheimer's disease show numerous cognitive and memory impairments. As a result of recent advances in genetic engineering techniques, an ever-increasing number of genetically modified mice reported to model Alzheimer's disease are being developed. Before these mice can be used to test possible treatments for Alzheimer's disease, the validity of these models must be tested by assessing the symptoms they exhibit. Behavioural tests of working and long-term memory in mice are available, however, the majority of these test rely upon visual tasks, while rodents are primarily olfactory animals. The use of two olfactometer tests, an odour discrimination task and a matching to sample task, are proposed as assessments of long-term and working memory, respectively, which would be able to take advantage of the mouse's superior sense of smell. I will report the results of pilot studies using C57Bl/6J mice and outline the proposed experiments using two background strains used in Alzheimer' mouse models.

T9) Investigation of Cytoskeletal Proteins in Neurons of the Cat Lateral Geniculate Nucleus

Emily LeDue, Nathan Crowder & Kevin Duffy

Cytoskeletal elements such as neurofilament and the actin-binding protein spectrin maintain the structural integrity of neurons. The largest neurons in the cat dorsal lateral geniculate nucleus (dLGN), which compose the Y-cell class, are distinguishable by their inclusion of the heavy neurofilament subunit. This observation raises the possibility that other classes of dLGN neurons can also be distinguished by their cytoskeletal content. In this study of the dLGN we examined the relationship between cell size and cytoskeletal composition. We determined that neurons labeled for all neurofilament subunits were significantly larger than the general population, and that spectrin labeling was not linked to cell size, as it was evident in almost all dLGN neurons. Double labeling for the inhibitory neurotransmitter, GABA, revealed that neurons reactive for GABA very rarely exhibited labeling for neurofilament, indicating that neurofilament is predominant in or exclusive to projection neurons.

T10) Can You Tell if Kids Have Pain or Not? Detecting Deception in Children's Facial Expression of Pain

Katelynn Boerner, Christine Chambers, Ken Craig & Rebecca Pillai-Riddell

In pediatric pain assessment, facial expressions are a primary source of information when making inferences about a child's pain. While most facial expressions appear spontaneous, children are capable faking or suppressing displays of pain, thereby biasing ratings. The present study will examine the accuracy of health professionals and parents in detecting deception in children's facial expressions of pain. Health professionals (pediatricians, pediatric nurses, and pediatric physiotherapists), and parents will view 48 randomized video clips of children in four pain conditions (faked, genuine, suppressed, and neutral) and will judge which condition the child was experiencing, specify which cues they used in their decision, rate the child's pain, and rate their degree of confidence in their decisions. Data collection for this research is currently underway and preliminary findings from participants collected to date will be presented.

T11) Physiological Arousal in Children and Adults with an Autism Spectrum Disorder: Reliability, Individual Differences, and the Role of Context

Dörthe Haacker, Shannon Johnson, Vickie Armstrong, Susan Bryson & Jillian Filliter

A disturbance in the autonomic arousal level is a significant symptom in individuals with autism (Bryson, 2004). Whereas some studies have found elevated levels of arousal others report low levels of arousal (Musurlian, 1995; van Engeland et al., 1991). The dysfunction in the modulation of arousal might be related to core features of autism such as emotional dysregulation, orienting problems and gaze aversion. Those and other repetitive and self-stimulatory behavioural pattern are viewed as attempts to modulate and regulate arousal.

We will measure skin conductance responses, heart rate and respiration rate of children and adolescence with and without autism. Tasks will include basic orientation and habituation tasks in which conditions of novelty, context and predictability versus unpredictability are presented.

We expect to find atypical levels of arousal that distinguish individuals with autism from typical controls (i.e. longer duration to reach habituation, higher responsiveness to novel stimuli and strength of orienting response). Furthermore, we expect that arousal patterns will be related to temperament profiles and autistic characteristics reported by parents.

T12) Speed-Accuracy Tradeoff in Stroop Interference

Virginia Palango & Jason Ivanoff

Stroop interference reflects the effect of word reading on color identification. Scholars have typically measured the Stroop effect using response time as a dependant variable. Here, we assess the temporal dynamics of the Stroop effect by manipulating processing time and measuring accuracy using speed-accuracy tradeoff (SAT) methodology. Consistent with previous findings, an analysis of the RT distribution demonstrated that the Stroop effect increased with processing time. Interestingly, the SAT analysis revealed no significant Stroop effect. These results have a number of implications pertaining to the sensitivity of SAT methodology, the disparity between the Stroop effect and other compatibility effects, and they suggest that caution be taken when interpreting the timecourse of cognitive processing with temporal measures.

T13) Effects of Training on a Dual n-Back Task on the Components of Working Memory and Fluid Intelligence

Andrew Clouter & Gail Eskes

Working memory ("WM") is the ability to temporarily hold and work with information in mind in order to apply it to complex tasks. WM function has a number of components, including two systems that temporarily hold and rehearse verbal/auditory (phonological loop) and visual information (visuo-spatial sketchpad).

The capacity of these temporary storage systems can be measured with working memory span tasks that require an individual to hold increasing amounts of information in mind, such as digit or visual span, operation span or reading span. When task demands are novel or complex such as those that require attention switching, inhibition, dividing attention or goal maintenance, a third component is involved, called the central executive or supervisory controller.

The goal of the current study is to further investigate how the different components of working memory respond to training using a dual n-back procedure, and how this training might generalize to related cognitive functions, such as fluid intelligence.

T14) The Emergence of Imitation: Preliminary Findings from a Prospective Study of Younger Siblings of Children with Autistic Spectrum Disorders

Ainsley Boudreau, Isabel Smith, J. Brian, Susan Bryson, W. Roberts, C Roncadin, P. Szatmari & L. Zwaigenbaum

Imitation deficits are among the earliest predictors of autism by 12 months of age (Zwaigenbaum et al., 2005). However, little is known about the emergence of imitation in typical and atypical populations. The study prospectively examines the emergence of imitation in the first year of life in infants at high risk (siblings of children with autism) and low risk (no family history of autism) for receiving a later diagnosis of autism. First, we will replicate Nichols' (2005) findings demonstrating a fixed sequence of approximations to imitation in low risk infants. Next, we will extend the findings to a high-risk infant sibling sample using a novel detailed coding scheme. Initial evidence supports the use of a hierarchy of approximations to imitation as an approach to studying the emergence of imitation skills. Preliminary findings in overall imitation ability and approximations to imitation at 9 months of age will also be reviewed.

T15) Repent Your Statistical Sins

Mike Lawrence

Come exorcise your demons in this discussion of lamentably common misunderstanding and misapplication of statistical tools in psychology.

T16) Immune Activation Influences Agonistic Behaviour in the Texas Field Cricket *Gryllus texensis*

Evan Fairn & Shelley Adamo

Immune system activation can influence investment in reproduction. Specifically, individuals mounting an immune response can (1) invest more in reproduction to compensate for a potential reduction in future reproduction ('terminal investment') or (2) invest less in reproduction due to a trade-off with immunity. Here we test whether mounting an immune response affects investment in a behaviour associated with reproductive success (agonistic behaviour) in males of the field cricket *Gryllus texensis*. Immune-challenged males were less aggressive than controls 5 minutes post-treatment but not at 90 minutes or 24 hours post-treatment. Immune-challenged males won significantly less than half of fights against controls at 90 minutes post-treatment but not 5 minutes or 24 hours post-treatment. Our results suggest a potential trade-off between immunity and reproduction in this species.

T17) The Remarkable Story of Captain Trevor Greene

Ryan D'Arcy, Stephen Lindsay, Debbie Lepore & Trevor Greene

Out of respect, Captain Greene took his helmet off when sitting down with the Afghan people in 2006. He was attacked from behind with an axe blow into the top of his head. Thought to be dead, no. Well then vegetative, no. Well then immobile for life, no. Captain Greene and his partner Debbie vowed he will walk again. In the summer of 2010, he stood alongside Debbie during their wedding vows. Functional MRI has tracked the reorganization of Captain Greene's motor regions over the last year - 4 years after the injury.

T18) Repertoire Composition, Song Structure and Singing Behaviour in the Hermit Thrush

Sean Roach, Lynn Johnson & Leslie Phillmore

The primary objective of this study is to provide a thorough description of repertoire composition and singing behaviour in the hermit thrush (*Catharus guttatus*), heretofore unavailable in birdsong literature, as well as to compare song aspects between geographically distinct populations. We recorded spontaneously singing territorial males during the breeding season in Halifax and Maine, and then compiled song type repertoires, measured song type characteristics, and analyzed song syntax. Overall, individual song type repertoires were organized in a similar manner between populations, although there were some geographical differences with regard to structural characteristics (e.g., note duration) of song types. There was no evidence of song type overlap within or between populations; that is, repertoires were individually distinct. There were no differences in song syntax between populations, with all individuals singing with immediate variety and exhibiting a moderate level of stereotypy (neither 100% predictable nor completely random).

T19) Effects of Gambling-Related Cues on Automatic Activation of Gambling Outcome Expectancies

Melissa Stewart & Sherry Stewart

Research on outcome expectancies has been extremely influential in the field of alcohol addiction. Despite the theoretical significance of outcome expectancies in the field of addictions, a paucity of research has investigated the impact of outcome expectancies on gambling behaviour. The current study aims to investigate factors that facilitate the activation of gambling outcome expectancies using both response time (RT) and self-report modes of assessment. Specifically, the present research will assess whether the presentation of gambling-related cues leads to the automatic activation of outcome expectancies stored in gamblers' memory networks. We will also examine how the activation of positive outcome expectancies may be substantially enhanced by gambling-related cues among those with increased problem gambling severity. Gambling outcome expectancies will be assessed by having gamblers (N = 120) complete an outcome expectancy RT task, as well as self-report measures of gambling outcome expectancies. Gambling behaviour outcomes will be assessed via self-report.

T20) A Checklist for Documenting Laboratory Factors Affecting the Behavioral Phenotyping of Mice

Richard Brown, Timothy O'Leary, Rhian Gunn & Heather Schellinck

Laboratory factors have a significant influence on the behavioral phenotyping of mouse models (Schellinck, H.M., et al. 2010. Adv.Study Behav., 41, 255-366). Despite attempts to control such effects, they persist. Since the behavioural phenotype is the product of both genetic and environmental (epigenetic) factors, the laboratory environment will always be a factor in determining the behavioral phenotype of mouse models. Our aim in this presentation is to create a checklist of laboratory variables that can influence behavioral results. We examine four categories of confounding variables: the mouse itself, the housing environment, the test environment, and the experimenter. Such a checklist will serve a number of functions. First, it will alert researchers to "hidden" environmental variables that can influence behavior; second, it will provide an outline for writing informative methods sections; third, it will provide a procedure for discussing result; and fourth, it will provide a procedure for comparing results across different laboratories. Such a checklist may also be published as supplemental data to behavioral phenotyping papers. This is the first draft of such a checklist, which we plan to revise and publish.

T21) Learning, Memory and Social Behaviour in the 3xTG-AD Mouse Model of Alzheimer's Disease

Kurt Stover & Richard Brown

The 3xTG-AD mouse model of Alzheimer's disease (AD) is a transgenic mouse possessing three human transgenes with mutations implicated in AD encoding for APPswe, PS1M146V, and tauP301L. These mice develop extracellular amyloid beta plaques as well as neurofibrillary tangles at around six months of age and deficits in learning and memory starting at four months of age. However mice from different colonies may have different behavioural phenotypes. Published studies have used mice from the University of California, Irvine (UCI). We obtained mice from Jackson Laboratories and will test them at two, six, and twelve months of age on a behavioural test battery to assess their learning, memory, and social behaviour and compare our results with those of mice bred at UCI. We have now tested mice at two months of age and found deficits in learning in the Morris water maze and better performance on the rotarod compared to wildtype controls.

T22) The Perfectionism Model of Binge Eating: A Self-Perpetuating Cycle

Sean Mackinnon, Simon Sherry, Aislin Graham, Sherry Stewart, Daba Sherry, Stephanie Allen, Skye Fitzpatrick & Dan McGrath

Why do people binge eat? Theory suggests binge eating is a cyclical behaviour maintained by interpersonal, affective, and biological processes. The perfectionism model of binge eating (PMOBE) is a theoretical model which clarifies the processes underlying this cyclical behaviour. Specifically, concern over mistakes is seen as a destructive aspect of perfectionistic personality which contributes to a cycle of binge eating through four maintenance variables: interpersonal discrepancies, low interpersonal esteem, depressive affect, and dietary restraint. This research represents an empirical test of this theoretical model in an undergraduate sample (N = 200) using a three-wave longitudinal design. Data were analyzed using path analysis and results provided consistent support for the PMOBE. Supplementary analyses also supported the incremental validity and the specificity of this model. The PMOBE offers a framework for understanding how key contributors to binge eating work together to generate and to maintain binge eating.

T23) The Relationship between Empathy and the Visual Scanning of Emotional Faces

Dana Higginbotham, Shannon Johnson, Jillian Filliter & Heath Matheson

Research suggests that individuals with less empathic abilities have greater difficulty in tasks of facial expression recognition compared to those with higher empathic abilities. Within individuals with Autism Spectrum Disorder (ASD), the relationship between empathy and behavioural performance on face processing tasks can partly be accounted for by how individuals with ASD are visually scanning faces. Research has yet to examine whether scan patterns can also account for the differences in tasks of facial expression recognition between high and low empathic individuals without ASD. The present study examined the relationship between empathic characteristics and efficiency in visually scanning emotional faces among typically developing individuals. The results of this study suggest that less empathic individuals are spending similar amounts of time scanning informative facial regions and more time scanning uninformative facial regions. The present study was the first to examine the relationship between empathy and the visual scanning of emotional faces within a typically developing population.

T24) Making Working Memory Better: Training the Central Executive

Ryan Wilson, Stefan Allen & Gail Eskes

This study examined the locus of training effects in Working Memory (WM) performance as measured in the n-back task. We hypothesized that training on the n-back task would target domain general mechanisms and thus training in one domain (e.g., verbal) would generalize to the other domain (e.g., spatial). Four groups of healthy young adults trained on either verbal or spatial tasks. Subjects also completed verbal and spatial tests at both 0- and 2-back load at baseline and at post-training after 3 weekly training sessions via a website interface. Spatial training significantly reduced RT on both verbal and spatial 2-back tasks, while verbal training reduced RT on only the verbal task. Both Verbal and Spatial trainers were more accurate for the verbal task and more accurate for the 0- than the 2-back task. These results some domain specificity, at least in early training.

T25) Working Working Memory

Stefan Allen, Ryan Wilson & Gail Eskes

Working memory (WM) forms the scaffolding for much of cognitive behaviour and is often affected in neurological disease, thus effective treatments are needed. Working memory consists of a central executive (CE) that directs two maintenance buffers (verbal and spatial). Practice on WM tasks improves performance, but the locus and thus generalizability of improvements in working memory capacity are unclear. The purpose of this study was to examine the locus of training effects (i.e., maintenance buffer vs CE) in WM performance as measured in the n-back task. In addition the Sternberg task was included as a transfer task to further examine generalizabiliy. Eighteen healthy young adults were randomly assigned for 3 days of training to a control group (non-specific training), spatial training group (practice only on spatial n-back) and verbal training group (practice on verbal n-back). It was hypothesized that training on the n-back will improve CE performance, and therefore performance benefits (e.g., reaction time and % accuracy) will generalize between domains (e.g., spatial & verbal) and further generalize to an untrained (transfer task) WM task, the Sternberg. Overall, results suggest domain transfer (from verbal to spatial and vv), but transfer to another working memory test was not seen. The locus of training effects will be discussed.

T26) The Effects of Repeated Stress in Adolescence on Sensorimotor Gating, Affect, and Memory in Male and Female Rats

Namrata Joshi & Tara Perrot

Repeated stress causes enduring effects when present during development, especially to brain regions under construction. I wanted to investigate whether repeated cat odour stress during adolescence results in enduring, sex-specific changes in prefrontal cortex-mediated sensorimotor gating, hippocampal function and measures of anxiety-like and depression-like behaviour in rats. Prepulse inhibition (PPI) of the acoustic startle response was used as a measure of sensorimotor gating. Performance in the elevated plus maze (EPM) and open field (OF) tests were used to determine levels of anxiety-like behaviour. The appearance of anhedonia (a symptom of Major Depression) was assessed using the sucrose preference test (SPT) while hippocampal function was measured using a novel object recognition (NOR) test. The results obtained so far reveal clear sex differences in responding, with adult female rats showing a greater effect of the repeated stressor exposure suggesting that females may show greater vulnerability to the impact of adolescent stress.

T27) There is a Coyote in my Backyard!

Simon Gadbois

On October 27th 2009, Taylor Mitchell was killed by coyotes in the Cape Breton Highlands National Park. This was only the second fatal attack on a human on record, and the first on an adult. I was involved within weeks in the investigation of what may have happened. For the past 18 months I have been consulting with Parks Canada and talking to biologists at the Department of Natural Resources in the hope of finding a permanent solution to the current problem, real or perceived, with our local coyote population. I will discuss the known issues, the perceived ones, and the potential solutions, to deal with problem animals, or populations more likely to interact with humans.

T28) Examining the Video Deficit Effect in 12 – 42 Month Olds

Oriane Landry & the Students of PSYO 3091

Imitation is an important mechanism in social learning that can aid a child acquire new knowledge and skills. In the present study, twenty-nine infants and toddlers ranging in age from 12-42 months were assigned to five different groups depending upon age, and presented with a scripted imitation game, in both a live and video model, in which target words and actions were examined. Results revealed that at 12-18 months, children display little to no imitation of either words or actions in the live and video models. As age increases, children begin to improve their imitational skills, however an increase in performance of imitation in the video model requires a higher age level than the live model, as a video deficit is displayed. Performance of imitational skills of both actions and words in children in the 36-42 month-old group were just as well in the video condition as the live condition.

T29) Characterizing the Hemodynamic Response Function of White Matter Using 4T fMRI

Leanne Fraser, T. Stevens, C. Liu & Ryan D'Arcy

The idea of fMRI activation in white matter is controversial. However, recent research has shown reliable activation in the corpus callosum using tasks designed to elicit interhemispheric activation. Despite advancements in the detection of fMRI activation in white matter, all calculations and analyses of white matter data have been based on the assumption that the hemodynamic response function (HRF) of white matter is the same as that found in gray matter.

We presented word and face stimuli to the visual hemifields. The task used a modified version in which single events were included to isolate the HRF. Functional MRI data were collected at 4 T, using an asymmetric spin echo spiral sequence. It was possible to extract time course data for isolated events. White matter responses were analyzed and compared to those in gray matter (parietal lobe, cingulate, visual cortex). Characterization and interpretation of the results will be presented.

T30) Investigating Sex Differences in Face Recognition and Gaze Direction

Sarah Whitzman, Laura Goodman, Heather Phelan & Shannon Johnson

A robust finding in the face processing literature is that faces with direct gaze are more readily recognized than faces with averted gaze'. A recent study in the Johnson lab found that males exhibited better recognition for direct relative to averted gaze faces but that females did not. To further examine this sex difference, the current study examined gaze direction and face recognition in 40 (20 males and 20 females) healthy adults, using a computerized face recognition task. We also investigated the relationships between gaze direction and factors that have been shown to be important when processing facial stimuli (i.e., attractiveness, familiarity, distinctiveness, and androgyny). Overall, both male and female participants demonstrated poor face recognition regardless of the sex of the face stimulus. However, we found that both males and females rated direct gaze faces as more attractive, more familiar, and more masculine than faces with averted gaze; this has implication for future face processing research.

T31) Factors Influencing Card-Sorting Performance in 3-year-olds

Denise Lewis, Meghan Tower & Oriane Landry

The Dimensional Change Card Sort (DCCS) is a widely used task assessing cognitive flexibility in preschoolers. The task is a simplistic adaptation of the Wisconsin Card Sort Task in which children are presented with two targets and two possible sorting dimensions, and are explicitly told on every trial how to sort the cards (by shape or by colour). Nevertheless, 3-year-olds are notorious for successfully sorting in one dimension but perseverating when instructed to change dimensions. We will report on two experiments examining potential factors in 3-year-olds' performance: instruction and repetition.

T32) It's All in the Expression: Cricket Tales

Russell Easy & Shelley Adamo

Many of the processes insects require to overcome an immune response involve the work of several genes and subsequently proteins. Transferrin is an iron-binding protein that is considered an acute phase protein and is upregulated in response to a pathogen stress. Nitrous oxide is a signaling and effector molecule involved in the insect response to pathogen challenge. The prophenoloxidase-activating system is an essential component of the insect immune response with a role in wound healing and melanotic encapsulation. The ability to survive an immune challenge may be directly dependant on upregulation of these compounds. Using real-time PCR we are examining how these compounds change in response to both stress and immune challenge. Once we have identified changes in expression, we will explore how stress influences the expression of immune-related genes in response to an immune challenge.

T33) Put on a Happy Face! Relating Emotional Competence and Emotion Regulation

Amanda Hudson & Sophie Jacques

Emotion regulation (ER) - the ability to exert control over emotional displays - may depend on several related skills. The current study aimed to decompose ER in 5- to 7-year-olds into its theoretical constituents. Children's responses to receiving a disappointing gift and parent-reported ER were used as indices of ER. Predictor variables included neutral and affective inhibitory control, neutral and affective theory of mind, emotion knowledge, display rule knowledge, language, and parent-reported social disposition and emotionality. ER following receipt of the gift was related to age, affective inhibitory control, theory of mind, and display rule knowledge. Parent-reported ER was related to social disposition and emotionality. After controlling for all other variables, affective inhibitory control and aspects of display rule knowledge uniquely predicted ER in the disappointing gift paradigm. These findings suggest that ER requires 1) *knowledge* of contexts requiring ER and 2) *inhibitory* skills required to execute ER.

T34) Introducing an Endogenous Attention Systems Test

Mike Lawrence & Shannon Johnson

The Attention Network Test (ANT) has become a popular research tool for quantifying several phenomena of attention. However, non-optimal design of this test limit it's power and interpretability. This talk will present a new attention test, designed to obviate the limitations of the ANT and extend its measurement to further phenomena of attention. Pilot experiments will be presented that validate its composition and applicability to both basic and clinical research.

T35) Human Sensitivity to Auditory Temporal Irregularity

Moragh Jang, Rachel Dingle, Susan Hall & Dennis Phillips

The human auditory system is extraordinarily sensitive to temporal periodicities in sounds, and to differences in the periodicities in sounds. What is far less clear is human sensitivity to deviations from perfect periodicity. This is important, because the failure of the auditory system to encode periodicities (e.g., in listeners with central auditory pathologies) is devastating perceptually, and experimental "jitters" of stimulus time structure in normal listeners is deleterious to speech perception. Detection thresholds for this kind of temporal jitter have not previously been presented. We report an experiment in which we directly measure sensitivity to the temporal irregularity of otherwise perfectly periodic click trains. Our data suggest that thresholds for the detection of temporal irregularity are not those expected from a simple Weber's Law, and that two perceptual operations may be contributing.

T36) Sensitivity to Morphological and Orthographic Stress Cues When Reading Multisyllabic English Words

Erin Sparks & Helene Deacon

In English words, certain letter patterns are associated with stress placement (e.g., *-age*; trochaic stress), and research suggests that adults are sensitive to these stress cues. Our study investigates whether this sensitivity differs when a cue is orthographic vs. morphological, and links stress cue sensitivity to reading ability. 104 participants completed a lexical decision task; its items were grouped into sets ending in the same letter pattern. These endings represent either a(n) orthographic cue, consistent with stress (*COURage*); orthographic cue, inconsistent with stress (*corSAGE*); or morphological cue, consistent with stress (*SHORTage*). 40 of the participants also completed measures of word reading and morphological awareness. We expect fewer errors when an orthographic cue is consistent vs. inconsistent, fewer errors for morphological vs. orthographic cues, and that the magnitude of the latter effect will be related to morphological awareness. Moreover, we expect better readers to be more sensitive to all stress cues.

T37) Visual and Auditory Factors in Sound Localization Acuity

Dennis Phillips, Chelsea Quinlan & Rachel Dingle

Historically, sound localization acuity was thought to be correlated with head size. Heffner and her colleagues (1992-2007) have dispelled this notion by showing that sound localization acuity is highly (and inversely) correlated with the species' width of field of best vision, and that when this factor is partialed out, the relation of acuity to head size vanishes. Her hypothesis is that one function of the auditory system is to provide source location data to the visual system so that the latter can bring the field of best vision into alignment with the auditory target. Thus, if one has a narrow width of field of best vision, then one must be a highly accurate sound localizer to achieve that end. But through what mechanism would this evolutionary pressure exert its effect on sound localization acuity? This talk provides one answer to that question.

T38) Exploring Attention in Item-Method Directed Forgetting

Kate Thompson & Tracy Taylor

In the item-method directed forgetting paradigm, participants are presented with study items one at a time, which are each followed by an instruction to either remember or forget. Typically, a directed forgetting effect is seen, where subsequent memory is better for R items compared to F items. Interestingly, research suggests that the allocation of attention differs after R and F memory instructions, as well. For example, Taylor (2005) found a greater magnitude of IOR (longer reaction times to targets presented in the same, rather than a different location than the word) after F instructions compared to R instructions. The present experiments used similar attentional paradigms to continue investigating the attentional effects of directed forgetting in order to determine when they occur, what their effects are, and attempt to understand what their purpose might be.

T39) The Relationship Between Inflammation and Neurotrophin Function in Depression

Cai Song

Monoamine deficits, hyper-HPA axis, inflammation and neurotrophin dysfunction may contribute to the etiology of depression. Most proinflammatory cytokines can activate the HPA axis and reduce monoamine syntheses. However, the relationship between inflammation and neurotrophic system, and between the HPA axis and neurotrophic system is unknown. In a depression model, olfactory bulbectomized rats, the microglia were activated, which was correlated with increased concentrations of proinflammatory cytokines interleukin (IL)-1, IL-6, tumor necrosis factor-a, and interferon-g in the hippocampus, and elevated CRF mRNA expressions in the hypothalamus. However, the expressions of brain-derived neurotrophic factor and nerve growth factor were down-regulated. TrK receptors were also down-regulated, whereas receptor p75 was up-regulated in the the model when compared to controls. An omega-3 fatty acid reduced inflammation and the HPA activity, normalized neurotrophic system and improved animal behavior. These results suggest that inflammation may contribute to the dysfunction of the HPA axis and neurotrophic system in depression.

T40) Is tobacco addiction nicotine addiction?

Sean Barrett

Both the positive (e.g. incentive motivation) and negative (e.g. withdrawal relief) reinforcing properties of tobacco smoking are often attributed to a single factor: nicotine. However, there is surprisingly little unequivocal evidence that nicotine per se has either negative or positive reinforcing properties in the absence of other pharmacological (e.g. psychoactive effects of non-nicotine tobacco ingredients) and non-pharmacological factors (e.g. expectancy effects; sensory-motor aspects of tobacco use). In a recent series of studies, my research group has found that many of the negative reinforcing properties of smoking appear to be attributable to non-nicotine tobacco constituents and/or the sensory-motor aspects of smoking, while many of tobacco's positive reinforcing properties appear to be attributable, at least in part, to psychological factors such as expectancy. In this presentation the roles of nicotine and various non-nicotine smoking factors for tobacco addiction will be discussed.

T41) Spatial and Temporal Dynamics of Visual Attention

Yoko Ishigami & Raymond Klein

Snyder (1972), using visual search, and McLean et al. (1973), using rapid serial visual presentation (RSVP), examined illusory conjunctions of features (Treisman & Gelade, 1980) in the domains of space and time, respectively. Seeking to develop a methodology for exploring whether "slippage" (failure to bind features) in space and time are related, the purpose of our pilot study was to replicate the methods and findings of these two classic studies. Participants searched for a coloured letter presented in a circular array all at once (visual search) or in a rapidly presented stream of single letters presented at fixation (RSVP). After each trial in each paradigm, they performed two tasks: identifying and localizing (in either space or time, depending on the paradigm) the target letter. Replicating previous findings, errors came predominantly from items adjacent to targets. Details of error patterns will be described and implications of future studies will be discussed.

T42) Validating a Measure of Parental Sleep Beliefs in the School-Aged Population

Meredith Bessey, Aimee Coulombe & Penny Corkum

Sleep problems are commonly reported in typically developing children, and are even more prevalent in clinical populations, such as children with ADHD and autism. Often times sleep problems in children go untreated, and parental beliefs about sleep may be a contributing factor. Developing a measure of these beliefs is an important first step in adapting and modifying these beliefs in order to improve treatment utilization and adherence. A newly developed sleep beliefs questionnaire was administered to parents of typically developing children (n=179) and of children with ADHD and autism (n=176). The goal of the current study was to begin to test the validity and reliability of the measure. A four factor structure emerged accounting for approximately 50% of the variance. The results indicate promise for this measure to be used in both typically developing and special needs populations, as a way to explore parents' beliefs about sleep and sleep problems in their children.

T43) Development of Intrinsic Neurons in the Mushroom Bodies of *Drosophila melanogaster*

Katie Goodine, Nancy Butcher & Ian Meinertzhagen

In insects, the mushroom bodies are paired, higher-order integrative brain structures implicated in olfactory learning and memory, courtship behaviors, spatial learning, and context generalization in visual learning. Extensive research on the anatomy and function of the mushroom bodies in the ant and honeybee has been conducted, and there is now growing interest in the mushroom bodies of the fruit fly *Drosophila melanogaster* because of its relatively simple organization and genetic manipulability. We investigated the use of genetic tools to label cells of the main input region of the mushroom bodies, the calyx, and to identify the number of mushroom body intrinsic neurons, called Kenyon cells, at three different developmental stages; the first prepupal stage (P0%), the eighth phanerocephalic pupal stage (P50%), and the freshly emerged adult stage (P100%). The implications of our findings will be discussed.

T44) The Effects of Alcohol on Smoking Cue Reactivity

Marcel Peloquin

Alcohol appears to alter smoking behaviour through physiological and psychological means. To tease these factors apart, experimental manipulation is required to understand the motivational and physiological roles alcohol has on tobacco use. 30 male and 30 female non-dependent smokers who are moderate drinkers will be assigned to an alcohol, placebo group, or a control group. They will watch four 2-minute videos, two during the ascending limb of alcohol absorption, and another two during the descending limb. Half the participants will watch smoking cues during the ascending limb and the other half during the second video of the descending limb. After each video presentation, participants will rate their desire to smoke on both the positively reinforcing and the negatively reinforcing dimensions of tobacco on questionnaires. This research will help determine if differences exist between psychological and physiological aspects of alcohol on smoking cues.

T45) Autism as a Cognitive Style: Visual Orienting and the Broader Autism Phenotype

Oriane Landry

Autism is characterized by severe impairments in social communication and behaviour. In my model of autism, later emerging impairments are a function of early emerging differences in attention, how and where infants direct their attention. There is compelling evidence that autism is caused by genetic factors; not only is there a higher incidence of autism within families, but there are also numerous autistic-like traits exhibited by non-impaired family members, referred to as the Broader Autism Phenotype (BAP). I will present recent data examining the relationship of the BAP to visual orienting.

T46) Testing the Embodied Account of Object Recognition

Heath Matheson & Patricia McMullen

Mounting evidence demonstrates that passively viewing images of tools both potentiates motor responses and activates frontoparietal networks responsible for programming action. To account for these findings, embodied theories of cognition propose that the same sensory-motor neural networks are involved in encoding and retrieving object knowledge. We investigated whether motor programs play a necessary role in the retrieval of object knowledge. Participants performed an object naming task while squeezing a sponge with either their right or left hand. The objects were either man-made (e.g. hammer) or natural (e.g. horse). We hypothesized that, if activation of motor programs are necessary to retrieve object knowledge, this arbitrary motor act should interfere with object naming, especially for categories that are associated with motor acts. Across two experiments we show that sensory-motor simulations are involved in the retrieval of both man-made and natural objects, suggesting that both are embodied.

T47) Sensory and Motor Mechanisms of Oculomotor Inhibition of Return

Zhiguo Wang, Jason Satel & Raymond Klein

We propose two mechanisms underlying inhibition of return (IOR) in the oculomotor system: sensory and motor. Sensory mechanism: Repeated visual stimulation results in a reduction of visual input to the superior colliculus (SC); consequently, saccades to targets which appear at previously stimulated retinotopic locations will have longer latencies than those which appear at unstimulated locations. Motor mechanism: The execution of a saccade results in asymmetric activation in the SC; as a result, saccades which reverse vectors will have longer latencies than those which repeat vectors. In the IOR literature, these two mechanisms correspond to IOR effects observed following covert exogenous orienting and overt orienting in the absence of peripheral stimulation, respectively. We predict that these two independent mechanisms will have additive effects; a prediction that is confirmed in a behavioral experiment. We then discuss how our theory and findings relate to the IOR literature.

T48) Sleep Architecture in Children with ADHD and Their Typically Developing Peers

Andre Benoit & Penny Corkum

Over the last 30 years, there have been highly inconsistent findings across studies which used polysomnography (PSG) to compare sleep architecture profiles of children with ADHD to their typically developing (TD) peers (e.g., Sadeh, 2007). These inconsistent findings could be related to methodological issues (e.g., participant medication status, co-morbidities). Therefore, the current study used PSG technology to analyze sleep architecture variables of a medication-naïve, rigorously diagnosed sample of children aged 6 – 12 years with ADHD (n=15), compared to a sex- and age-matched sample of their TD peers (n=15). Multivariate analysis (MANOVA) revealed significant differences between ADHD and TD groups in terms of sleep architecture. Specifically, children in the ADHD group had a significantly longer latency to REM sleep, significantly fewer REM cycles, and less Stage 1 sleep than TD peers. Also, children in the ADHD sample had significantly longer sleep onset latencies, but significantly shorter sleep duration than TD peers.

T49) Odour Matching in Dogs: Applications and Challenges

Simon Gadbois, Kate Thompson, Brittany Hilton, John Christie & Vincent LoLordo

Odour matching is used with forensic canines (dogs used as «expert witnesses») and wildlife conservation dogs (to match individuals from scats). Unfortunately, we believe that the most common technique (even standardized in Europe), the scent line-up, is fundamentally flawed as it makes a perceptually difficult task even more challenging by adding an unnecessary element of memory. In the last year we have been investigating a number of other approaches including «same/different» responses to pairs of odours, and short delayed-matching-to-sample procedures. A comparison of all three approaches and the discussion of a true perceptual challenge (odour detection in matrices) will be presented.

T50) The Role of Faces in Item-Method Directed Forgetting

Chelsea Quinlan & Tracy Taylor

This study examined whether or not the specialized system used to process, encode, and recognize faces contributes to memory processes, such as the ability to remember and intentionally forget. Each experiment in this study consisted of an item-method directed forgetting paradigm. In the study phase, facial expressions were presented one at a time, each followed by either an instruction to Remember or an instruction to Forget. Following the presentation of all items in the study phase, memory performance was measured using a yes/no recognition test. Although a directed forgetting effect (greater memory performance for Remember-cued items than Forget-cued items) was found for Neutral facial expressions across all of the experiments, this was not the case for emotional facial expressions (Angry and Happy facial expressions). Critically, it was found that there was a link between timing and the directed forgetting effect for emotional facial expressions such that when the stimulus presentation duration was decreased (i.e., really short), there was no effect of emotional valence on directed forgetting; however, when the inter-trial interval was increased (i.e., really long), there was an effect of emotional valence (both negative and positive valence) on directed forgetting.

T51) Wait a Minute, Forget That: Using Natural Speech Cues to Induce Directed Forgetting
Ben Corrigan, Jonathan Fawcett & Tracy Helmick

To examine directed forgetting due to natural speech cues, participants were asked to rate audio presentations for their quality. The audio clips included mistakes (Errors), which were acknowledged (F instruction) and then corrected (Correction), as well as other items (Reference). During a surprise two-alternative forced choice recognition task, participant's accurate recognition for Correction and Reference items was significantly higher than for Error (F) items. It is hypothesized that because participants were not intentionally attempting to memorize the material, the forget instruction initiated a cognitive process that withdrew attention from the Error items.

T52) Drinking Motives, Alcohol Use, and Alcohol-Related Problems: A Cross-National Comparison and Validation of the Drinking Motives Questionnaire-Revised Among College Students
Marie-Eve Couture, Sherry Stewart, E. Kuntsche, M. Cooper & Roisin O'Connor

The aims of this study were to explore cross-national similarities and differences in the structure of the Drinking Motives Questionnaire-Revised (DMQ-R), in the mean endorsement levels of social, enhancement, coping, and conformity drinking motives, and in the links between drinking motives and both heavy drinking and alcohol-related problems among college students. Confirmatory factor analysis, analysis of variance, and structural equation modeling were performed on data from Canada, the United States, Hungary, Spain, the United Kingdom, Israel, Ireland, and Brazil. Results indicate that the structure of the DMQ-R did not consistently yield adequate fit to the data across countries; however, the four-factor structure of the measure was superior to an alternate three-factor structure across countries. Further similar drinking motives predicted heavy drinking and alcohol-related problems across countries. Similarities were also found in the rank order of drinking motive endorsement across countries but some cross-national differences emerged in the mean levels of endorsement of particular motives.

T53) Aladdin's Cave: New Lamps for Old in Illuminating the Tiny World of the Fly's Eye
Steve Shaw

The question of the origin of the several electrical responses recorded extracellularly from the eyes of flies has gained prominence with the adoption of this cellular model for molecular genetic studies of neural transmission, in *Drosophila*. In particular, the origins and dynamics of the electroretinogram's ON- and OFF-responses are not understood and have been widely misinterpreted. An accidental discovery here has suggested that the usual method of free-field illumination is of limited utility in exposing the properties of the ON and OFF signals. Introducing a new method of illumination to limit these deficiencies has allowed separation of the ON-OFF system from the more elemental responses of the photoreceptors and monopolar neurons of the first synaptic zone, exposing these clearly for the first time in the electroretinogram. Currents from different monopolar neuron do not interact and as expected, sum independently and linearly. In contrast, the ON-OFF system collects light from many facets across the eye in an apparently cooperative, non-linear summative fashion.

T54) The Effects of Prenatal Psychological Stress and Postnatal Environmental Enrichment on the Development of Anxiety and Depressive Behaviours and Neural Changes in the HPA Axis

Amanda Green, Kal Mungovan, Meaghan MacLean & Tara Perrot

We investigated the impact of a novel prenatal psychological stress (PPS) on sex-specific ontogeny of behaviours related to social interaction, anxiety and depression, and the role of postnatal environmental enrichment (EE) to prevent/rescue these changes. Activity was decreased in juvenile EE offspring compared to standard cage. Juvenile females showed greater anxiety-like behaviour than males and EE removed this difference. Juveniles exposed to EE also showed less depressive-like and antisocial behaviour, although antisocial behaviour increased in males with exposure to both PPS and EE. In adulthood, PPS females displayed more depressive-like behaviour compared to all other groups, and this correlated with their juvenile levels of depressive-like behaviour. Interestingly, there were sex-specific effects of PPS and EE on CRH-ir cells in the PVN of the hypothalamus, with both PPS and EE affecting males to a greater extent. The implications of these findings will be discussed.

T55) Reading for Meaning: Investigating Children's Sensitivity to Morphological Structure During Contextual Reading

Kyle Levesque & Hélène Deacon

Children utilize multiple avenues of information in order to accurately read and understand written words. One such avenue is presumed to be the morphological structure of complex words (e.g., *bold* + *ly*). In particular, the frequency of the base morpheme (e.g., *bold* in *boldly*) appears to have a privileged role in children's reading. Complex words with high frequency bases (e.g., *questionable*) are read more accurately and understood to a greater extent than words with low frequency bases (e.g., *conceivable*).

The current investigation stems from the fact that children appear to be sensitive to the morphological structure of words presented in isolation. Yet, it remains unclear to what extent the effects of base frequency are influenced by contextual reading. Thus, the purpose of this experiment is to explore the relationship between base frequency (high vs. low) and context (isolation vs. sentence) by examining their combined effect on word reading and comprehension.

T56) White Matter Connectivity in Bilaterally Deaf Individuals vs. Hearing Controls

Therese Chevalier, Aaron Newman, Vanessa Lim, Heather Maessen & Manohar Bance

In the event of sensory deprivation, such as hearing loss, cross modal plasticity may occur as brain areas are reallocated for other functions. Previous studies have reported functional differences, such as auditory areas responding to visual stimuli, as well as differences in diffusivity along white matter pathways between deaf and control participants. We used Diffusion Tensor Imaging (DTI) to infer the integrity and orientation of white matter tracts noninvasively and *in vivo*. Tractography was conducted from regions of interest, which included the primary auditory cortex, planum temporale, and V5 of the visual cortex in both hemispheres. Significant differences in connectivity were found between V5 and the planum temporale in the left hemisphere in deaf patients as compared to controls. Similar differences were noted at trend levels in the right hemisphere. These findings support the hypothesis that structural changes occur to facilitate cross modal cortical reorganization in response to sensory deprivation.

T57) Motivation and Data Quality Across the Term: How to Get Cleaner Data in Just Five Minutes

Jonathan Fawcett & John Christie

We present a questionnaire currently in development to measure attitudes towards research following completion of a variety of experimental tasks. The questionnaire itself is available in both computerized and paper form and requires between two and five minutes to complete. It is our prediction that our questionnaire could be developed to allow researchers to predict the quality of the data a subject will return, thus allowing researchers to account for that in their models. Analyses of the pilot project we ran this year will be presented. We are actively seeking research laboratories in our department willing to help recruit subjects for this project.

T58) The Effects of Eating Dennettia tripetala fruit on locomotor and anxiety-related behaviours in CD1 mice

Daniel Ikpi, Richard Brown & Sunday Bisong

Dennettia tripetala fruit induces a mild stimulating effect when ingested; it is used to spice foods in Southern Nigeria. It is believed to reduce social anxiety and increase sociability. The present study investigated the stimulatory effects of D. tripetala on locomotor and anxiety-like behaviours in CD1 mice in the open field, elevated plus maze and light-dark transition box. Thirty singly housed male CD1 mice were divided into 3 groups of 10. The Control group were fed normal rodent pellets. The Low-dose group were fed a 10% D. tripetala diet (10g of D. tripetala mixed with 90g of rodent pellets). The High-dose group were fed 15% D. tripetala diet (15g of D. tripetala mixed with 85g of rodent pellets) daily for 4 weeks before testing. Both doses of D. tripetala reduced the number of line crosses and the frequency of centre square entries in the open field while the high dose reduced the number of line crosses in the light-dark box and the frequency of rearing in the open field. The low dose of D. tripetala increased the time in the open arms in the elevated plus maze, while the high dose decreased the number of stretch attend postures in the light-dark box. The results indicate that chronic ingestion of D. tripetalareduced locomotor and exploratory behaviour in CD1 mice at both doses while the low dose may have an anxiolytic effect. Future studies will examine the effect of D. tripetalaon motor learning, motor control and sociability.

P1) Personality Factors and Susceptibility to Burnout in Nova Scotia Ground Ambulance Paramedics: A Cross-Sectional Study

Zach Dewar, Alix Carter & Gail Eskes

Burnout Syndrome is prevalent in human service professions but little research has been done on causes of burnout in paramedics. Burnout can compromise provider and patient safety, and can have long-term health consequences. Research demonstrates that personality traits of providers can serve as protective or risk factors in the development of burnout syndrome. This study aimed to examine the association of personality traits with risk of burnout in paramedics, in an effort to better understand burnout syndrome in paramedicine. We used an online survey to ask paramedics in Nova Scotia about their work, burnout and personality. We then analyzed their results using linear regression, and found that neuroticism was a risk factor for burnout, with higher age, calls/month and years of experience contributing as other risk factors. These findings are consistent with previous research on other professions and allow for the tentative generalization of that research to paramedic populations.

P2) headgames.org: A Secure Platform for Web-Based Experiments

Mike Lawrence & John Christie

With the growing prevalence of personal computers and internet connectivity, web-based experimentation has become an increasingly enticing modality for psychological research. However, there as yet exists no unified solution for recruitment, participation tracking, and provision of compensation. We have therefore developed headgames.org, a website dedicated to connecting researchers and everyday web users, facilitating the conduct of web-based experimentation. Specific features include demographically targeted recruitment, secure and confidential data transfer, and integration of a trusted solution for provision of financial compensation (PayPal). We invite proposals from the local community for projects to post during the now begun beta phase of headgames.org.

P3) Epigenetics and the Future of Psychology and Neuroscience

Richard Brown

The genomes of humans, rats, mice and fruit flies are now known and the effects of manipulating known genes on brain and behaviour are being investigated. Because environmental variables also modulate brain function and behaviour, the study of how epigenetic variables modulate the expression of genes regulating brain function and behaviour is important for the future of psychology and neuroscience. The action of environmental factors on neurochemical systems and intracellular transduction pathways of neurons alters the structure of histone proteins in the chromatin of the cell nucleus and these histone proteins regulate gene expression. Thus, epigenetic processes provide the mechanism for environmental regulation of gene expression and the solution to the nature-nurture problem. This poster describes the importance of epigenetic approaches to the study of neural and psychological processes involved in behavioural development, aging, learning, memory, social behaviour and neuropsychiatric disorders.

P4) A Neurodevelopmental Profile of the 3xTg-AD Mouse Model of Alzheimer's Disease

Caitlin Blaney and Richard Brown

The 3xTg-AD mouse model of Alzheimer's disease (AD) is a transgenic mouse possessing three human transgenes with mutations implicated in AD encoding for APPswe, PS1M146V, and tauP301L (Oddo S, et al., 2003, Neuron, 39, 409-421). These mice develop extracellular amyloid beta plaques as well as neurofibrillary tangles at around six months of age and deficits in learning and memory starting at four months of age. Because maternal behaviour can have long-lasting effects on offspring behaviour (Weaver IC, et al., 2004, Nat Neurosci. 7, 847-854), we used a cross-fostering design to examine the interaction of maternal genotype and pup genotype on neurobehavioural development from 2-24 days of age. The 3xTg pups showed earlier eye opening, pinnae detachment and earlier development of the hind limb grasp reflex and rooting reflexes than the wild-type (WT) controls. The 3xTg mothers had lower levels of maternal behaviour than the WT mothers. Although WT mothers showed more maternal behaviour than 3xTg mothers, WT pups reared with 3xTg mothers showed advanced development of the rooting reflex, hindlimb grasp reflex and loss of the cross-extensor reflex than when raised by WT mothers. These results suggest that there is a complex interaction between maternal care and pup genotype in the rate of early pup development of these mice. We are rearing these mice to adulthood to determine the relationship between maternal behaviour, early developmental parameters and the development of Alzheimer-like behavioural disruptions in the aging 3xTg-AD model mice.

P5) Orientation Specificity of Contrast Adaptation in Rodent Primary Visual Cortex

Aaron Stroud, Emily LeDue & Nathan Crowder

Previous studies examining orientation specificity of contrast adaptation in cats have demonstrated the presence of two main subsets of cells in primary visual cortex (V1): 1) Non-oriented adapting neurons, which adapt to stimuli of all orientations; and 2) Orientation-specific adapting neurons that only adapt to stimuli matching the cell's preferred orientation. It has been proposed that local intracortical connections between neighbouring V1 neurons may mediate this aspect of cortical contrast adaptation. Adaptation is thought to be related to a cell's position in the pinwheel orientation map of V1. Mice lack these orientation columns, thus it is expected that visually responsive cells within V1 will adapt to gratings at any orientation. We recorded from neurons in V1 of mice and compared the amount of adaptation produced by gratings at both preferred and orthogonal orientations. Both types of stimuli produced contrast adaptation in most cells. Implications for cortical contrast coding are considered.

P6) Is Cognitive Performance Associated with Preclinical Markers of Parkinson's Disease?

Laura Burns, Kim Good, John Fisk & Harold Robertson

Parkinson's disease (PD) is the second most common neurodegenerative disorder, after Alzheimer's disease. It is clinically characterized as a chronic motor disorder. However, non-motor disturbances such as impaired cognitive functioning also affect most patients, prior to treatment as well as throughout the duration of the illness. Specific deficits include: recall of verbal material, working memory, and processing speed. Often, these cognitive deficits are not clinically apparent, however, they have been detected in research settings. The current investigation uses a battery of neuropsychological tests, on twelve healthy controls and twelve patients with PD. An analysis of the differences between the groups will augment our existing research on olfactory deficits and neuroimaging biomarkers. We intend to further characterize the potential preclinical markers of PD so the future at-risk population can be diagnosed before the motor symptoms become apparent and the damage is extensive.

P7) Modeling Inhibition of Return as Short-Term Depression of Early Sensory Input to the Superior Colliculus

Jason Satel, Zhiguo Wang, Thomas Trappenberg & Raymond Klein

Inhibition of return (IOR) is an orienting phenomenon characterized by slower behavioral responses to spatially cued, relative to uncued targets, when the cue-target onset asynchronies (CTOAs) are long enough that cue-elicited attentional capture has dispersed. Here, we implement a short-term depression (STD) account of IOR within a neuroscientifically based dynamic neural field model (DNF) of the superior colliculus (SC). In addition to the prototypical findings in the cue-target paradigm (ie., the biphasic pattern of behavioral enhancement at short CTOAs and behavioral costs at long CTOAs), a variety of findings in the literature are generated with this model, including IOR in averaging saccades and the co-existence of IOR and endogenous orienting at the same location. Many findings that cannot be accommodated by this model could be accounted for by incorporating cortical contributions.

P8) The Effect of the Pulfrich Stereophenomenon on Reaching and Perceptual Judgment

Stéphanie Beckett & David Westwood

The resistance of action to perceptual illusions has been attributed to the use of stimuli whose illusory features result from processing beyond the divergence of the dorsal and ventral streams. If this is correct, then illusions originating from mechanisms occurring prior to V1 should affect action and perception similarly. One such illusion is the Pulfrich Stereophenomenon in which the illusion of depth is created when a moving stimulus is viewed with a neutral density filter (NDF) over one eye.

Participants (n=11) made perceptual distance judgments or reaching movements in response to a moving stimulus displayed in a virtual aiming environment with and without a NDF over their left eye.

The results showed a somewhat greater effect of the filter for action than perception, in contrast to most studies which show greater effects of illusion for perception than action. Implications for models of visual control of action and perception are discussed.

P9) Visuo-Spatial Learning and Memory in Six Mouse Models of Alzheimer Disease

Richard Brown, Rhian Gunn & Timothy O'Leary

The Morris Water maze (MWM) is the most commonly used test for visuo-spatial learning and memory in rodents and is frequently used to test cognitive deficits in mouse models of Alzheimer Disease (AD) (D'Hooge & De Deyn, 2001, Brain Res Rev 36, 60-90). We have tested males and females of five strains of AD model mice (APPInd, APPSwInd, AppSwe/PS1, AppSwe/PS1dE9, 5XFAD) and their wildtype (WT) control strains in the MWM between 3 & 24 months of age and are now testing the 3XTg strain. Both longitudinal and cross-sectional studies were completed. This paper summarizes the effects of genotype, age and sex on measures of learning (latency, search strategy) and memory (percent time in correct quadrant, annulus crossings) in the MWM and discusses background gene effects which influence performance independently of transgene effects. The most significant genotype effects were found in the APPInd and 5XFAD mice and there were few significant sex differences, although females appear to use a different search strategy than males. All strains showed reduced performance with age. Background genes for retinal degeneration and albinism affect performance independently of AD transgenes. The results demonstrate the use of the MWM as an analytical tool in dissociating the factors underlying visuo-spatial learning and memory in AD model mice.

P10) Neural and Behavioural Analyses of the 5XFAD Mouse model of Alzheimer's Disease

Dietmar Hestermann, Timothy O'Leary, Arjan Blokland & Richard Brown

In the current study the 5XFAD (B6SJL-Tg) mouse model and wild-type mice (N=25) will complete a behavioural test battery which will assess anxiety, locomotor activity, motor ability, visuo-spatial learning and memory, olfactory memory and visual ability. Following behavioural testing mice will be sacrificed and neural changes within brains will be assessed using histology techniques. Following research questions will be answered. What is the difference in behaviour, memory and cognitive function in 6-9 month transgenic mice compared to wild-type mice? This will be investigated by using behavioural tests. How is the neural structure affected in 6-9 month old transgenic mice? Which brain areas important for visuo-spatial learning and memory differ? Histology techniques are used to answer these questions. Of special interest is the amyloid plaque load resulting from elevated beta amyloid levels and changes in cell number indicating neurodegeneration. Areas of interest are the hippocampal formation and cortex.

P11) Exploring the Modulation of Attentional Capture by Spatial Attentional Control Settings: Converging Evidence from Event-Related Potentials

Yoko Ishigami, Jeff Hamm, Jason Satel & Raymond Klein

Attentional control settings (ACSs) are typically defined by a single target-relevant location. The purpose of the current study is to seek converging evidence from ERPs for a spatial ACS defined by multiple target-relevant locations, using the methods in Ishigami et al. (2008). Any one of four figure-8s brightened uninformatively (cue) before presentation of a digit target, calling for a speeded identification. A spatial ACS was encouraged because in different blocks the digit targets appeared only on the horizontal or vertical axis. Performance was more impaired following the attended invalid cues than following unattended invalid cues. The amplitudes of early cue-elicited visual evoked potentials (VEPs) were greater when the location the cue was presented in was target-relevant than when the location was target-irrelevant. Thus, both performance and ERP showed evidence of successful spatial ACSs defined by multiple target locations.

P12) Antidepressants as Preventive Analgesics for Chronic Post-Surgical Pain

Jean Liu, Jana Sawynok, Gary Allen and Allison Reid

Up to 50% of post-operative patients may later develop chronic pain due to inadvertent nerve injury at the surgical site. While the extent of the pain experienced varies depending on the surgical procedure, the high incidence of chronic pain has given rise to the need for preventive treatments. Recent rodent experiments in our laboratory have shown that amitriptyline, an antidepressant, can prevent the long-term development of post-surgical pain if it is administered at a critical time in relation to a surgically-induced nerve injury. Since amitriptyline prevents noradrenaline and serotonin reuptake, we predict that the efficacy of peri-operative administration of the drug may be due to the modulation of descending noradrenergic and serotonergic pain inhibitory systems. We are currently comparing three different nerve injury models in order to determine whether changes occur in the descending noradrenergic pathways that could constitute a neural substrate for the action of amitriptyline.

P13) Modeling Mechanisms of Reversed Effects of Manipulability on Naming and Categorizing Objects

Joshua Salmon, Patricia McMullen & Thomas Trappenberg

Our behavioural research has shown that manipulable objects are named more quickly and categorized more slowly than non-manipulable objects (Salmon & McMullen, 2009). This pattern of effects was explored using extensions of a computational model of Yoon, Heinke and Humphreys' (2002) Naming Action Model (NAM) by determining the success of three possible mechanisms to simulate it. The first mechanism was based on semantic organization of the system (crowded versus sparse domains or semantic branching) whereby crowded categories (those consisting of many objects) were postulated to be categorized more quickly and named more slowly than sparse categories. The second mechanism was more perceptual, and based on the input of similar objects whereby highly similar objects were postulated to be categorized more quickly and named more slowly (note that - similarity could refer to either visual similarity or the similarity between appropriate action responses: action similarity). The third mechanism was based on the effect of a dedicated Action Module postulated in the NAM model to be active for manipulable but not for non-manipulable objects. This Action Module was postulated to speed the naming of manipulable objects, but slow their categorization. Our computational simulation supported a 'reversal effect' due to the first two mechanisms, but not the third. Hence, a sparse versus crowded semantic architecture and similarity of the input objects could account for the 'reversal effect' but a dedicated Action Module could not. These results suggest that actions associated with objects can affect their identification at specific and non-specific levels without appealing to a module dedicated to coding these associations.

P14) The Localization of Neurofilament Subunits in the Rat Primary Somatosensory Cortex Throughout Postnatal Development

Lin Lin Ngu, Song, Katharine Cheung & Kevin Duffy

Neurofilament is the predominant shape-providing intermediate filament in neurons. Within the visual system neurofilament has been linked to particular cell types and functions. In primate visual cortex the protein is localized most heavily within cells that compose the regions outlining metabolically distinct 'blobs'. In this study we examined whether the non-uniform distribution of cell characteristics and physiological functions within the rat somatosensory system is linked to the localization of neurofilament subunits. In adult rats we observed neurofilament-positive neurons preferentially located outside of metabolically distinct 'barrels'. Investigation of neurofilament subunits throughout development revealed that for two subunits (medium and heavy) reactivity was initially located most densely within processes that formed barrels. These results demonstrate a distinction very early in postnatal development between the labeling patterns of neurofilament subunits that reveal either cells indigenous to the cortex (light) or processes whose is likely thalamic (medium and heavy).